

Zon Kitiyakara

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SUMMARY

Electrical Engineering student with hands-on experience in **embedded systems, sensor integration, and digital logic**. Skilled in C/C++, MSP432, Arduino, and hardware debugging. Strong ability to design and validate embedded hardware through real-time testing and iterative optimization.

EDUCATION

San Jose State University – Charles W. Davidson College of Engineering

Bachelor of Science in **Electrical Engineering** | Expected December 2026

Dean's Scholar Award, 2024

Relevant Coursework: Microprocessor-Based System Design, Embedded Systems Laboratory(TI-RSLK / MSP432 Programming), Digital Logic Circuit Design, Circuits and Systems

SKILLS

- **Programming & Firmware:** C, C++, MSP432 (ARM Cortex-M4), Arduino, millis() timing, interrupt-driven logic
- **Embedded Hardware:** ADCs, GPIO/PWM, SPI/I²C/UART, sensor calibration, ultrasonic rangefinding
- **Digital Design:** Verilog, FSM design, RTL implementation, testbenches, Vivado simulation, FPGA prototyping
- **Testing & Tools:** Oscilloscope, debugging, threshold tuning, prototype testing, CCS, LTSpice, Git/GitHub

ELECTRICAL PROJECTS

Smart Proximity Sensor System – Arduino Uno Microcontroller

Personal Project | Fall 2025

- Designed an ultrasonic proximity detection system with LED/buzzer alerts, improving accuracy **40%** through filtering and sampling optimization.
- Currently adding a **180° servo-scanning mechanism** for angle-based mapping and 3× wider coverage.
- Developing an **OLED UI** with multi-mode control (Silent, Stealth, Aggressive) using I²C and state-machine firmware.
- Building modular firmware to support upcoming Bluetooth telemetry, data logging, and low-power modes.

IR Distance Classification System - TI-RSLK Maze Lab | MSP432 LaunchPad

San Jose State University | EE Project | Fall 2025

- Programmed 14-bit ADC data acquisition and nonlinear calibration for Sharp IR sensors.
- Implemented real-time state logic (8 path scenarios & 7 safety conditions) for automated navigation and detection.
- Integrated sensor feedback systems demonstrating reliable signal processing and real-time response design.

Virtual Pet Simulator – Arduino-Based System

Personal Project | April 2025 – May 2025

- Built a fully interactive **C++ UI** using dual-button input for real-time display control.
- Developed **bitmap-animated OLED graphics** with live system-state tracking for hunger, sleep, and happiness levels.
- Eliminated blocking delays by implementing **non-blocking FSM-style logic** using millis(), achieving <5 ms input latency.

Moore FSM Sequence Detector - Verilog & Nexys FPGA Board

San Jose State University | EE Project | Spring 2025

- Designed a five-state Moore FSM in Verilog to detect the sequence “11011.” Created testbenches and validated behavior using Vivado simulations.
- Deployed on Artix-7 FPGA for hardware validation of digital logic behavior.

EXPERIENCE & LEADERSHIP

Covour Coffee - Palo Alto, CA

Barista | June 2025 – Present

- Supported high-volume operations while improving team workflow, communication, and customer experience.

Society of Asian Scientists and Engineers - San Jose, CA

Co-Event Coordinator | June 2023 – June 2024

- Organized an Intel-sponsored networking event attracting **100+ STEM students** from four universities.
- Increased participation **by 60%** through targeted outreach and coordinated event planning.
- Delivered presentations and facilitated discussions, strengthening **technical communication and leadership skills**.